TECHNOLOGICAL CHINA / PPT 6

CHINA 2030

Geoeconomics, artificial intelligence and technological competition

Collection synthesis · Scenarios and innovation frontiers

Sergio Gevatschnaider



The integrated thesis: competition requires whole systems

One model or company cannot explain the technology race.

01

Knowledge

Research, education and talent.

02

Compute

Chips, data centres and energy.

03

Industry

Suppliers, automation and scale.

04

Institutions

Finance, standards and coordination.

Ask who can manufacture, diffuse and sustain innovation alongside who invents.

Three measures, three different concepts

Indicators with different units and populations cannot be added.

2.80%

R&D expenditure as a share of GDP

China, 2025 · NBS

10th

Global Innovation Index

China, 2026 · WIPO

≈55%

Electric share of new car sales

China, 2025 · IEA

Capacity, ranking and adoption indicators measure different things.

SOURCE NBS China, R&D expenditure 2025 (Oct. 2026)

A map of the six presentations

From economic infrastructure to scenario analysis.

01 · Transformation

Economics, scale and learning

02 · AI

Models, agents and efficiency

03 · Chips

Semiconductors and compute

04 · Robotics

Control and embodied intelligence

05 · Vehicles and batteries

Charging, factories and value networks

06 · China 2030

Geoeconomics, scenarios and Latin America

Six complementary decks: each includes a laboratory and audiovisual resources.

TRANSFORMATION

01 · ECONOMIC TRANSFORMATION

Industrialisation, innovation and structural change



SERGIO GEVATSCHNAIDER

From factor-driven growth to productivity

A growth-accounting framework.

01

Capital

More investment does not guarantee sustained returns.

02

Labour

Demography and skills reshape supply.

03

TFP

Organisation, knowledge and competition matter.

04

Diffusion

AI benefits require actual adoption.

Productivity differs from patent counts and market size.

R&D: China’s measurable investment in 2025

Official statistics released in October 2026.

3.93

Trillion yuan in R&D expenditure

2025 · NBS

2.80%

R&D intensity relative to GDP

2025 · NBS

+8.2%

Annual change in R&D spending

2025 · NBS

Intensity is expenditure / GDP; assess output indicators alongside it.

SOURCE NBS China, R&D expenditure 2025 (Oct. 2026)

New quality productive forces: concept and limits

Examine the slogan as an economic hypothesis.

Potential mechanisms

- Automation and fewer errors.
- New product development.
- Reorganisation of production networks.
- AI adoption in services and industry.

Empirical questions

- Does sector productivity increase?
- Does the return on capital recover?
- Is innovation radical or incremental?
- Who captures innovation rents?

The concept becomes useful when translated into testable indicators.

Five-Year Plan 2026–2030: innovation objectives

Public targets and priorities rather than achieved outcomes.

01

Basic science

Expand the research frontier.

02

Critical technologies

Address technology dependencies.

03

Application

Industrialise R&D results.

04

Digitalisation

Combine models, chips, cloud and uses.

Distinguish policy intentions, implementation and evaluation after the fact.

SOURCE 15th Five-Year Plan, science and technology priorities 2026–2030

Rebalancing growth: consumption and investment

China’s model faces domestic and external constraints.

Supply-side drivers

- Advanced manufacturing and exports.
- Investment in technology and infrastructure.
- Potential AI productivity gains.
- Economies of scale and learning.

Macroeconomic constraints

- Cautious private consumption.
- Property adjustment and household balance sheets.
- Marginal returns on investment.
- Ageing and the labour force.

Innovation must coexist with demand and capital adjustment.

SOURCE World Bank, China Economic Update (July 2026)

World Bank: a dated macroeconomic outlook

Institutional forecasts rather than historical outcomes.

4.4%

Projected real growth

China, 2026 · World Bank, July 2026

4.3%

Projected real growth

China, 2027 · World Bank, July 2026

Forecasts are revised; do not extrapolate them mechanically to 2030.

SOURCE World Bank, China Economic Update (July 2026)

INNOVATION

02 · INNOVATION ECOSYSTEMS

Science, platforms, business networks and geography



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WIPO 2026: performance and spatial concentration

Country rankings and clusters are different measures.

10th

China in the Global Innovation Index

139 economies · 2026

1st

Shenzhen–HK–Guangzhou cluster

Overall cluster ranking

25

Chinese clusters in the top 100

WIPO · 2026

Rankings are composite: examine methodology and economy size.

SOURCE WIPO, Innovation clusters 2026

Three Chinese clusters: complementary specialisations

There is no single homogeneous technology ecosystem.

01

Shenzhen

Hardware, electronics and rapid scaling.

02

Beijing

Universities, research and AI.

03

Shanghai–Suzhou

Advanced industry and regional networks.

Proximity reduces search costs and speeds up engineering iteration.

SOURCE WIPO, Innovation clusters 2026

From research to a competitive product

Technology transfer is not automatic.

01

Science

Theory, laboratories and talent.

→

02

Prototype

Testing and intellectual property.

→

03

Scaling

Capital, equipment and suppliers.

→

04

Market

Regulation, adoption and competition.

Industrialisation, rather than publication, may be the scarce link.

Patents and papers: valuable, partial evidence

The knowledge frontier is broader than counts.

What they contribute

- A map of inventive activity.
- Signals of research capabilities.
- Indicators of emerging sectors.
- Traces of collaboration networks.

What they cannot establish alone

- An invention’s profitability.
- Superior performance.
- Sustainable commercialisation.
- Economy-wide productivity.

Triangulate research, citations, products, adoption and value added.

Six capabilities and six vulnerabilities

A qualitative analytical synthesis rather than a score.

Talent

+ Scientific critical mass

– Uneven distribution

Compute

+ Data centre expansion

– Critical accelerators

Manufacturing

+ Broad supplier networks

– Specialised technologies

Batteries

+ Scale and chemistry

– Minerals and trade

AI

+ Competitive open-weight models

– Evaluation and safety

Market

+ Demand and platforms

– Pressure on margins

System competitiveness can vary considerably across stages.

CONVERGENCE

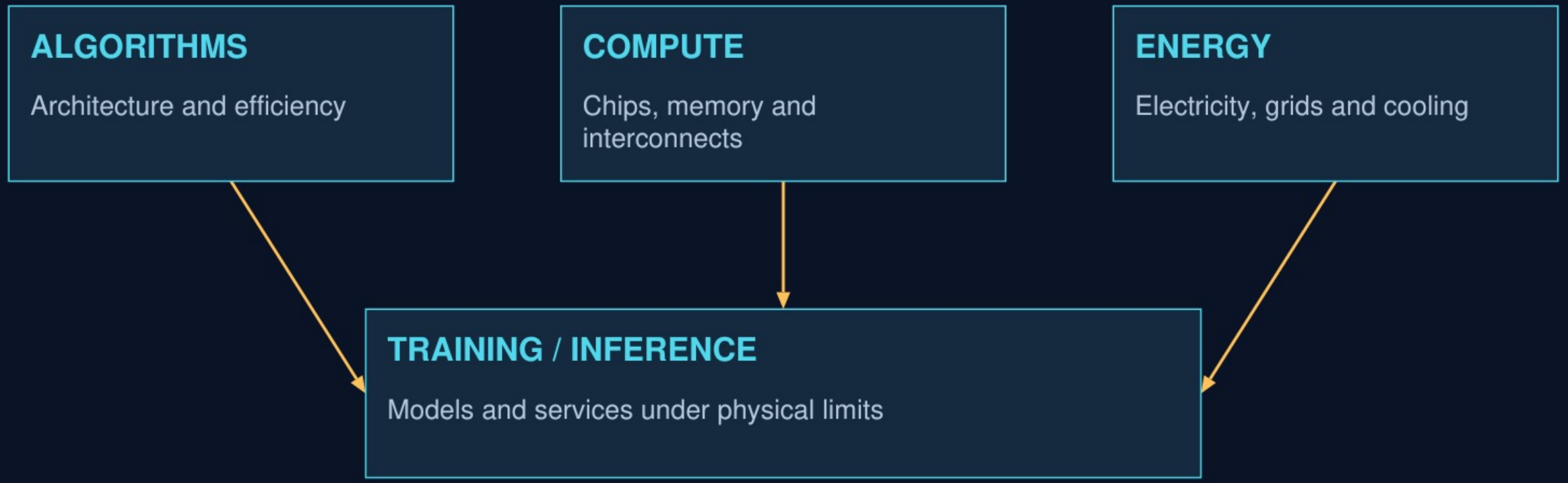
03 · TECHNOLOGICAL CONVERGENCE

AI, chips, energy, robotics and manufacturing



Complementary inputs in the AI economy

Distinguish training, inference and organisational adoption.



Adoption creates value when it integrates processes, people and validation.

Chinese AI: DeepSeek, Qwen and open-weight economics

Avoid comparing benchmarks under different conditions.

01

Costs

Training and inference efficiency.

02

Access

Open weights facilitate adaptation.

03

Agents

Tool use and workflows.

04

Evaluation

Safety, robustness and real performance.

Business adoption depends on quality, integration and operating risk.

The semiconductor bottleneck

Interdependencies beyond the lithography node.

Local capabilities

- Accelerator design.
- Systems and software engineering.
- Data centre infrastructure.
- Iterations in packaging and memory.

Persistent constraints

- Leading-edge lithography.
- HBM and packaging technologies.
- Critical tools and EDA.
- Interconnects and energy performance.

Autonomy is a vector of dependencies rather than a binary variable.

Robotics and embodied intelligence

From algorithms to physical performance.

01

Perception

Sensors, estimation and context.

02

Control

Safety, latency and robustness.

03

Data

Field experience and validation.

04

Production

Maintenance, costs and OEE.

A robot demonstration does not establish profitable deployment.

Electrification as an industrial platform

IEA shares for 2025: sales differ from component production.

≈55%

Electric share of China’s new car sales

2025 · BEV + PHEV

≈75%

Global electric car production

Manufactured in China · 2025

>80%

Global battery cell production

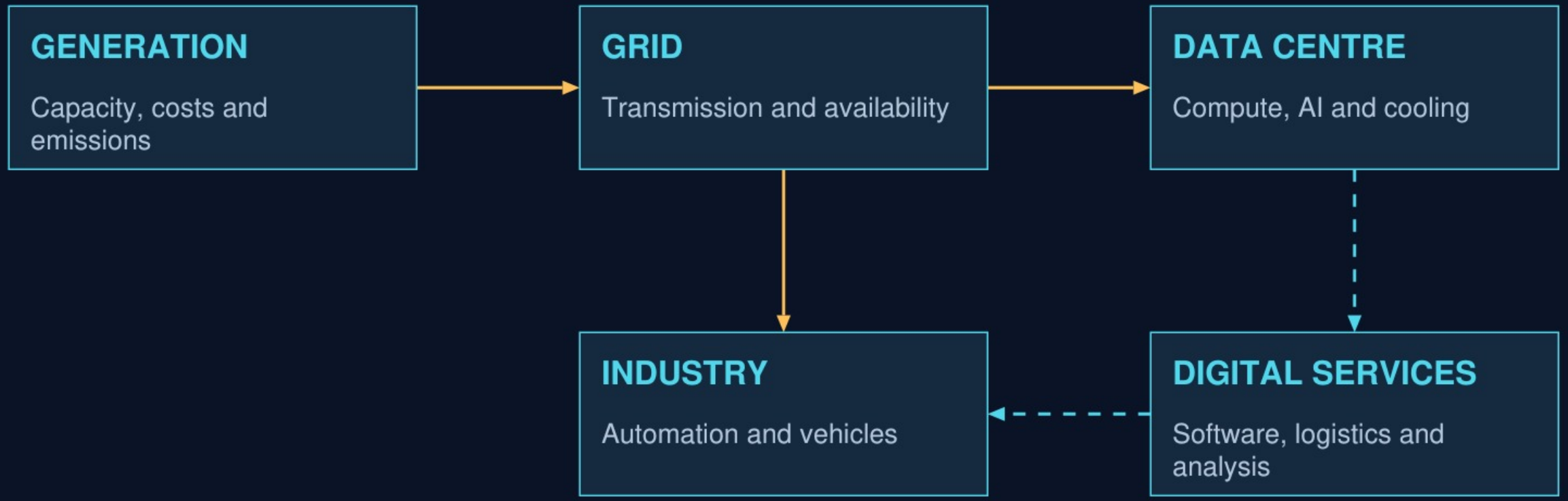
Manufactured in China · 2025

Production concentration can raise efficiency and exposure to shocks.

SOURCE IEA, Global EV Outlook 2026

Energy, compute and factories: a shared physical system

Electricity availability conditions the pace of adoption.



Gold: electricity. Dashed cyan: services and information.

GEOECONOMICS

04 · GEOECONOMICS AND DEPENDENCIES

Competition, interdependence and supply-chain reorganisation



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Four dimensions of technological competition

The problem goes beyond the bilateral relationship.

01

Design

Intellectual property and talent.

02

Manufacturing

Machinery, materials and scale.

03

Standards

Compatibility, certification and rules.

04

Markets

Prices, distribution and alliances.

Competition coexists with interdependence and diversification strategies.

Export controls: transmission channels

Effects depend on substitutes and adjustment times.

Direct effects

- Restricted access to certain equipment.
- Potentially dearer leading-edge compute.
- Changes in customers and suppliers.
- Redesign and substitution incentives.

Indirect effects

- Incentives for software efficiency.
- Investment may shift between stages.
- Greater trade uncertainty.
- Changes in scientific cooperation.

Evaluate effective capability and opportunity costs alongside policy announcements.

Regionalisation: decoupling or redundant networks?

Two dimensions: concentration and substitution.



Diversification does not guarantee resilience: common exposure matters.

Routes to technological internationalisation

Different strategies in response to trade barriers.

01

Exports

Final products and components.

02

FDI

Local plants and joint ventures.

03

Licensing

Technology, software and standards.

04

Services

Support, data and maintenance.

Entry strategy depends on the product, regulation and life cycle.

Excess capacity: a balanced economic reading

Distinguish productive investment, competition and allocation.

Possible benefits

- Lower prices for buyers.
- Industrial scale and learning.
- Faster technology diffusion.
- Competition and greater variety.

Risks and adjustments

- Excessively low margins.
- Underused assets.
- Financing and local debt.
- Trade conflicts and defensive measures.

Evaluate capital returns, utilisation and competitive conditions.

A sector dependency matrix towards 2030

A qualitative diagnostic framework without normative rankings.

AI

+ Models and software

– Inference costs

Chips

+ Local design

– Critical equipment

Robots

+ Industrial scale

– General-purpose robustness

Vehicles

+ Integration and volume

– Foreign tariffs

Batteries

+ Established chains

– Materials and recycling

Energy

+ Large investment

– Grids and storage

In dynamic scenarios one dependency may shrink while another emerges.

SCENARIOS

05 · CHINA TOWARDS 2030

Alternative scenarios, risks and sensitivity analysis



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Five scenarios for academic discussion

These are neither official forecasts nor estimated probabilities.

REFERENCE

Eight original model settings

AI BOOST

R&D 4.1%; talent 82; adoption 85

TRADE SHOCK

Chips 30; trade 30; energy 55

REBALANCING

Demand 90; coordination 82

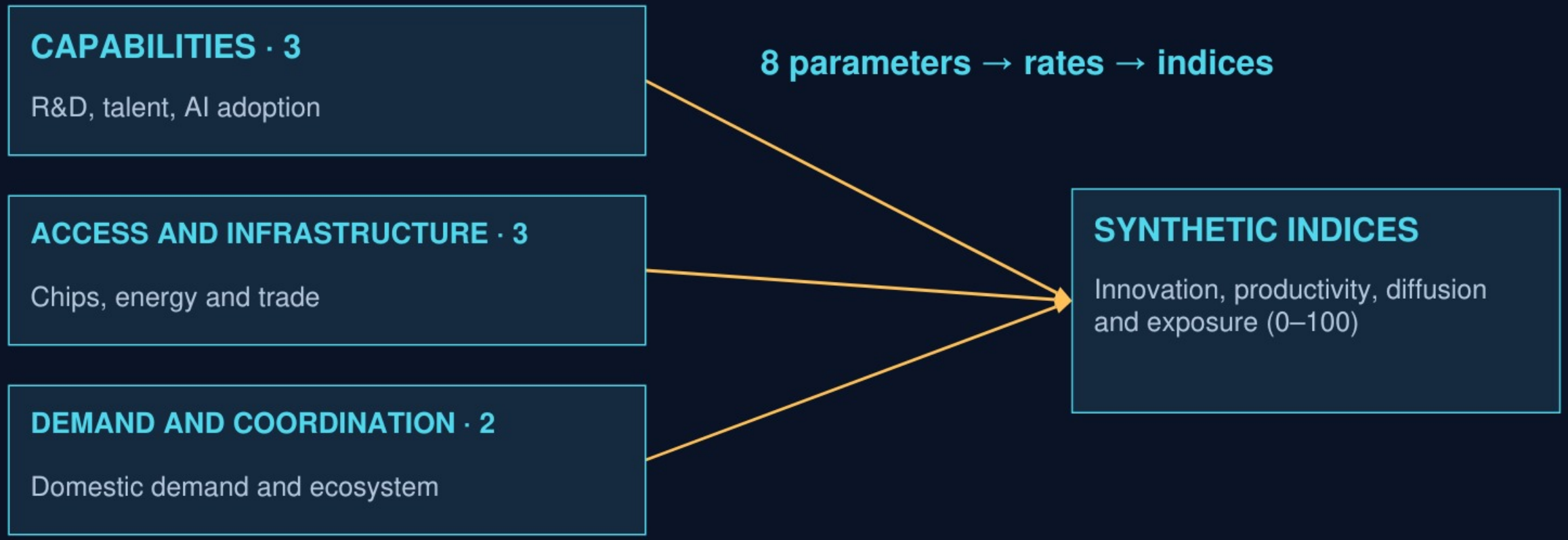
INTEGRATED STRATEGY

Joint improvements; R&D 4.2%

All five presets share equations and assign no probabilities.

The repository model’s critical variables

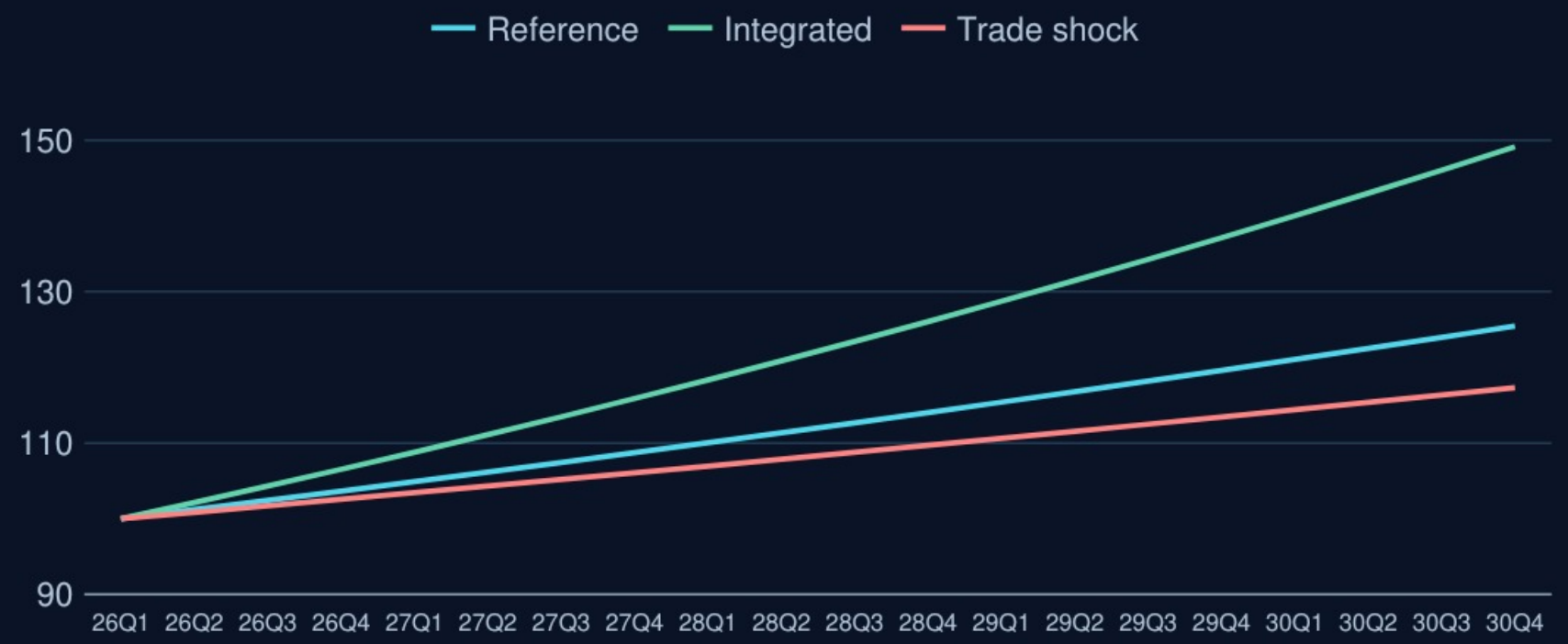
Parameters can be adjusted in the HTML laboratory.



Bounded additive rates: the model does not simulate hard bottlenecks.

Synthetic trajectories rather than observed projections

Innovation index · 2026 Q1 = 100.



Reference

125.4

Integrated

149.1

Trade shock

117.3

Same laboratory equations · 20 observations and 19 quarterly intervals.

Tail risks absent from an average scenario

Discrete technological shifts matter.

Negative shocks

- Disrupted critical imports.
- Persistent trade tensions.
- Slower domestic demand.
- Safety or infrastructure failures.

Positive shocks

- Algorithmic efficiency breakthroughs.
- Materials and battery innovation.
- Diffusion of agents across processes.
- Better ecosystem coordination.

Structural uncertainty cannot be reduced to a central estimate.

What our simulator measures and omits

Transparent and reproducible interpretation.

01

Measures

Sensitivity to explicit assumptions.

02

Compares

Scenarios and relative trajectories.

03

Does not estimate

Real GDP, probabilities or equity values.

04

Does not replace

Calibration, validation and microdata.

A useful teaching tool states the limits of its validity.

SOURCE Author's deterministic teaching model: uncalibrated indices, without predictive validity.

LATIN AMERICA

06 · IMPLICATIONS FOR LATIN AMERICA

Production integration, opportunities and policy choices



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Latin America: four transmission channels

Opportunities and risks vary by country and sector.

01

Trade

Minerals, capital goods and EVs.

02

Investment

Factories, services and logistics.

03

Technology

Licences, solutions and training.

04

Geopolitics

Diversification and standards.

Exposure to China combines external demand and domestic technology chains.

Value-added specialisation and extraction

Two strategies that can coexist under clear criteria.

Joining value chains

- Engineering and data services.
- Industrial components and testing.
- Maintenance and electronics.
- Energy solutions and software.

Avoiding passive dependencies

- Contracts without effective transfer.
- Excessive buyer concentration.
- Lack of standards and certification.
- No local supplier development.

Trade strategies are stronger when they build domestic capabilities.

An analytical agenda for firms and governments

From opportunity to investment assessment.

01

Map

Chains and local capabilities.

→

02

Compare

Costs, quality and standards.

→

03

Pilot

Measurable, reversible projects.

→

04

Scale

Audited results and financing.

The aim is to absorb transferable lessons from China.

Questions for a regional research agenda

Four avenues for academic projects.

01

Productivity

Where does AI raise TFP?

02

Dependency

Which networks cannot be substituted?

03

Trade

How do elasticities and prices change?

04

Institutions

Which policies show additionality?

Design causal identification before attributing changes to AI or China.

CONCLUSION

07 · SYNTHESIS AND FRONTIERS

Lessons from Chinese technological competition



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Five conclusions to close the collection

An integrated reading of all six presentations.

SCALE

Speeds learning; requires returns

CAPABILITIES

Connect research and production

INSTITUTIONS

Coordinate and evaluate outcomes

COMPETITION

Diffusion, innovation and margins

INTERDEPENDENCE

Efficiency and vulnerability coexist

An analytical synthesis requiring evidence and regular updating.

Open questions that will remain relevant in 2030

Prioritise good questions over absolute predictions.

Innovation capability

- Will R&D translate into productivity?
- Will compute gaps close?
- Will reliable agents and robots scale?
- Will investment be sustainable?

International economics

- How will global chains change?
- Will domestic demand rebalance?
- Which standards will emerge?
- What opportunities will Latin America have?

Technological China is an open transition with heterogeneous outcomes.

A critical reading guide: data, targets, scenarios and forecasts

Four categories that should remain distinct.

DATA

Observed or estimated result
Example: 2025 R&D

TARGET

Policy or company objective
Does not establish delivery

SCENARIO

Result conditional on assumptions
Example: laboratory presets

FORECAST

Estimate of future outcomes
Must state uncertainty

These are distinct evidence categories rather than a causal sequence.

Bibliography and sources for continued updating

A selection of primary sources, methodologies and statistics.

NBS
R&D expenditure 2025 · October 2026

WIPO
Global Innovation Index 2026 · methodology

WIPO
Innovation clusters 2026 · ranking

WORLD BANK
Rebalancing Growth · July 2026

IEA
Global EV Outlook 2026

IMF
China Article IV · February 2026

PLAN 2026–2030
Official science and technology priorities

TEACHING MODEL
Reproducible equations; uncalibrated

Full links in speaker notes and the bilingual web guide.

COLLECTION SYNTHESIS

China 2030: think in systems

Technology, productivity, institutions and interdependence

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